

Franziska Hoerbst

bouncing between research, art, activism & science communication



On the move



1998, Austria



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linktree

Research and education

Most recent

● The Unjournal

Evaluating research; to make *rigorous* research more *impactful* and *impactful* research more *rigorous*

2024 – 2025

Postdoc

● John Innes Centre

Computational Systems Biology

Statistical pitfalls in the sciences and what we can do to resolve them

2020 – 2024

PhD

● John Innes Centre & University of East Anglia

Computational Systems Biology

Doctoral Thesis supervised by Richard Morris

Bayesian Inference in the Analysis of RNA-Seq Data

2017 – 2020

BSc

● University of Vienna

Molecular Biology

Thesis in Computational Biophysics with Bojan Zagrovic at Max Perutz Labs

Exploring the Physicochemical Foundation behind Travelling Autologous Protein-mRNA Pairs in the Phloem of Plants

Science communication & teaching

2020 – 2022



more here

● 5MinutenClimateChance

Climate science podcast & radio broadcast

Co-founder, host, journalistic researcher & producer

→ Austrian SDG Award 2020

→ Austrian Environmental Journalism Prize 2020

→ Top 5 Salto Awards 2021

2022

Norwich, UK

● University teaching

Mathematics for Computer Scientists

Teaching assistant

2018 – 2020

Vienna, AT

● ScienceCenter Network Vienna

Hands-on science communication

in a job role called 'explainer', which is essentially a workshop host, who is developing hands-on and tinkering activities for audiences of all ages and backgrounds

is fascinated by

Life

Philosophy of science

Colours

Information Theory

Climate & Biodiversity Crises

Bayesian Statistics

The importance of cultural practices

Complex Systems

Slowing down & degrowth

Equality & Justice

The post carbon world

Nature & Humanity

New things every day

Research publications

- | | |
|-------------------------|--|
| 2022 | ● <u>Exact Bayesian inference for the detection of graft-mobile transcripts from sequencing data</u> |
| Royal Society Interface | F Hoerbst, M Tomkins, S Gupta, F Apelt, J Kehr, F Kragler, RJ Morris |
| 2023 | ● <u>baymobil: A Python package for detection of graft-mobile mRNA using exact Bayesian inference on RNA-Seq data</u> |
| Preprint | F Hoerbst, RJ Morris, M Tomkins |
| 2024 | ● <u>A Closed-Form Solution to the 2-Sample Problem for Quantifying Changes in Gene Expression using Bayes Factors</u> |
| Preprint | F Hoerbst, GS Sidhu, M Tomkins, RJ Morris |
| 2024 | ● <u>Re-analysis of mobile mRNA datasets highlights challenges in the detection of mobile transcripts from short-read RNA-Seq data</u> |
| Nature | P Paajanen, M Tomkins, F Hoerbst, R Veevers, M Heeney, HR Thomas, F Apelt, E Saplaoura, S Gupta, M Frank, DJ Walther, C Faulkner, J Kehr, F Kragler, RJ Morris |
| 2024 | ● <u>Bayesian Inference in the Analysis of RNA-Seq Data</u> |
| Thesis | F Hoerbst |
| 2025 | ● <u>On the overinterpretation of mass screening data – the example of mobile mRNA</u> |
| Trends in Plant Science | F Hoerbst, M Tomkins, RJ Morris |
| 2024 | ● <u>A Bayesian framework for ranking genes based on their statistical evidence for differential expression</u> |
| Under review | F Hoerbst, GS Sidhu, T Omori, M Tomkins, RJ Morris |
| 2024 | ● <u>What is a differentially expressed gene?</u> |
| Under review | F Hoerbst, GS Sidhu, M Tomkins, RJ Morris |

Further publications & engagement

- | | |
|---------------|--|
| 2024 | ● <u>Let's 'do different': Planetary Citizen Education Proposal</u> |
| Working paper | E Honeybun-Arnolda, P Stephanides, F Hoerbst, N Longhurst, D Maguire, H Hoechner |
| since 2020 | ● <u>5MinutenClimateChance</u> |
| Podcast | 100+ episodes filled with snack-bites of climate research |
| 2021 | ● <u>A day trip through phloem</u> |
| Blog | Easy read research project description |
| 2023 | ● <u>Life out there? Scanning plant space for new communication networks</u> |
| Blog | Easy read paper summary |
| 2023 | ● <u>Activism and Advocacy as a Researcher</u> |
| Panel | Panel discussion at the Annual Assembly of the Tyndall Centre for Climate Change Research |

finds balance in

- Music and silence
- Nature and the outdoors
- Arts and crafts
- Tech and low-tech
- Exploring movement and movements

speaks

- German and English
- Python fluently
- Cada día un pocito más español

loves speaking

- at international scientific conferences
- on panels (like previously on Women in STEM, [climate activism as researchers](#), ...)
- at public engagement events (like e.g. [Pint of science](#))
- with kind humans around a dinner table

the miscellaneous

2024

→ Best Presentation Award at MaxEnt24

Maximum Entropy (MaxEnt) is a valued, historical conference series in the field of Bayesian statistics where researchers from quantum physics and philosophy to economy come together once a year since to learn from each other about probability theory

2023

JIC Director Award

A yearly award given away by Director Graham Moore to individuals to acknowledge and honour outstanding scientific as well as cultural contributions to the John Innes Centre

since 2022

🌐 more here

• UEA Biodiversity and Climate Action Network

Particularly involved in the Education Working group, with the vision of changing all curricula towards engaging all students in concepts surrounding global citizenship

since 2022

• JIC Student Voice

Running campaigns to support the John Innes student community to thrive

2021

• Finalist DigiEduHack 2021

Climate Change Digital Education Hackathon

from 2012

• Radio Orange 94.0, Freies Radio Freistadt, & Radius 106.6

Speaker and producer

→ Student award by the Austrian Federal Ministry of Education and OE1 for best broadcast 2016

2017 & 2018

Coordination and execution of the charity radio marathons
4 Tage für den Sozialmarkt, 4 Tage für die Lebenshilfe

2016

→ Regional Movie Award @YOUKI Film Festival

2008 – 2017

Several accolades for music (Let's discuss the concept of music competitions, especially for kids and teenagers)

Contact for references

• Professor Richard J. Morris

Group leader at John Innes Centre

Primary PhD supervisor

Richard.Morris@jic.ac.uk

• Dr Noel Longhurst

Lecturer in Energy & Climate Change, School of Environmental Sciences, UEA and founding member of UEA BCAN

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